Ubuntu 64-bit - VMware Workstation 17 Player

Player

Activities Terminal Feb 10 00:52

vaidehi@ubuntu: ~

vaidehi@ubuntu: ~ vaidehi@ubuntu: ~

```
vaidehi@ubuntu:~$ gcc file_stat.c -o file_stat
gcc: error: file_stat.c: No such file or directory
gcc: fatal error: no input files
compilation terminated.
vaidehi@ubuntu:~$ ./file_stat
bash: ./file_stat: No such file or directory
vaidehi@ubuntu:~$ ^C
vaidehi@ubuntu:~$ ls
Desktop    Downloads  fork_exec.c  Pictures  Templates  Vaidehi
Documents  fork_exec  Music        Public    test.txt   Videos
vaidehi@ubuntu:~$ nano file_stat.c
vaidehi@ubuntu:~$ gcc file_stat.c -o file_stat
vaidehi@ubuntu:~$ ./file_stat file_stat.c
File Name: file_stat.c
File Size: 797 bytes
Number of Links: 1
File Inode: 264494
Last Access Time: Tue Feb 10 00:49:36 2026
Last Modification Time: Tue Feb 10 00:49:25 2026
Last Status Change Time: Tue Feb 10 00:49:25 2026
vaidehi@ubuntu:~$
```

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vaidehi@ubuntu: ~

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GNU nano 4.8 fork_exec.c

```c
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t pid = fork();

    if (pid == 0) {
        // Child process
        printf("Child PID: %d\n", getpid());
        execlp("ls", "ls", NULL);
        exit(0);
    } else {
        // Parent process
        wait(NULL);
        printf("Parent PID: %d\n", getpid());
    }
    return 0;
}
```

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Activities Terminal Feb 10 00:54

vaidehi@ubuntu: ~

vaidehi@ubuntu: ~ vaidehi@ubuntu: ~

GNU nano 4.8 file_stat.c

```c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/stat.h>

int main() {
    struct stat st;
    int fd = open("test.txt", O_RDONLY);

    if (fd == -1) {
        perror("open");
        return 1;
    }

    stat("test.txt", &st);
    printf("File size: %ld bytes\n", st.st_size);

    close(fd);
    return 0;
}
```

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Activities Terminal

```
vaidehi@ubuntu:~$ nano syscalls.c


Use "fg" to return to nano.

[1]+  Stopped                 nano syscalls.c
vaidehi@ubuntu:~$
vaidehi@ubuntu:~$ nano syscalls.c
vaidehi@ubuntu:~$ gcc syscalls.c -o syscalls
```

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Activities Terminal Feb 10 01:16

vaidehi@ubuntu: ~

```
vaidehi@ubuntu:~$ ./syscalls
Parent PID: 9727
Child PID: 9728
Desktop    Downloads  file_stat.c       file_stat.c.save.1  fork_exec.c  Pictures  syscalls    Templates  Vaidehi
Documents  file_stat  file_stat.c.save  fork_exec           Music        Public    syscalls.c  test.txt   Videos
Child finished
File Size: 13 bytes

Files in current directory:
syscalls
fork_exec.c
..
Vaidehi
.local
.
.cache
Documents
Pictures
Templates
Videos
.gnupg
syscalls.c
.config
file_stat.c
.bash_logout
.bashrc
fork_exec
Desktop
Music
test.txt
file_stat.c.save
.sudo_as_admin_successful
```

Activities Terminal vaic

```
Music
test.txt
file_stat.c.save
.sudo_as_admin_successful
Downloads
Public
.bash_history
file_stat.c.save.1
file_stat
.profile
vaidehi@ubuntu:~$
vaidehi@ubuntu:~$ touch file1.txt
vaidehi@ubuntu:~$ touch file2.txt
vaidehi@ubuntu:~$ chmod 777 file1.txt
vaidehi@ubuntu:~$ ls -l file1.txt
-rwxrwxrwx 1 vaidehi vaidehi 0 Feb 10 01:09 file1.txt
vaidehi@ubuntu:~$ sudo chown $USER file1.txt
[sudo] password for vaidehi:
vaidehi@ubuntu:~$ nano file1.txt
vaidehi@ubuntu:~$ nano file2.txt
vaidehi@ubuntu:~$ diff file1.txt file2.txt
1c1
< Hello World
---
> Hello Linux
```

Ubuntu 64-bit - VMware Workstation 17 Player

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Activities Terminal Feb 10 01:18

vaidehi@ubuntu: ~

```
> Hello Linux
vaidehi@ubuntu:~$ diff file1.txt file2.txt > mypatch.patch
vaidehi@ubuntu:~$ patch file1.txt mypatch.patch
patching file file1.txt
vaidehi@ubuntu:~$ history
    1  echo hello ubuntu
    2  date
    3  uptime
    4  time
    5  cal
    6  cat filename
    7  tty
    8  man command
    9  which command
   10  history
   11  id
   12  pwd
   13  whoami
   14  ls
   15  ping google.com
   16  ifconfig
   17  sudo app install net-tools
   18  pr file
   19  pr vaidehi
   20  lp file
   21  lpr file
   22  lpstat
   23  lpq
   24  lprm jobID
   25  cancel jobID
   26  mail
   27  sudo app install mailutils
```

```
44  sudo apt update
45  sudo apt install build-essential
46  nano fork_exec.c
47  gcc fork_exec.c -o fork_exec
48  ./fork_exec
49  gcc fork_exec.c -o fork_exec
50  ./fork_exec
51  touch test.txt
52  echo "Hello OS Lab" > test.txt
53  nano file_stat.c
54  nano syscalls.c
55  gcc syscalls.c -o syscalls
56  ./syscalls
57  touch file1.txt
58  touch file2.txt
59  chmod 777 file1.txt
60  ls -l file1.txt
61  sudo chown $USER file1.txt
62  nano file1.txt
63  nano file2.txt
64  diff file1.txt file2.txt
65  diff file1.txt file2.txt > mypatch.patch
66  patch file1.txt mypatch.patch
67  history
idehi@ubuntu:~$ ss
```